Shaft Head Frame

26/08/2019

Head Frame I

*Head frame can be made of wood, steel or concrete.

*Head frame with back legs.

*A type Head Frame.

*Four post head frame

*Six post head frame

*Tower Head frame.

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Head Frame Structure I

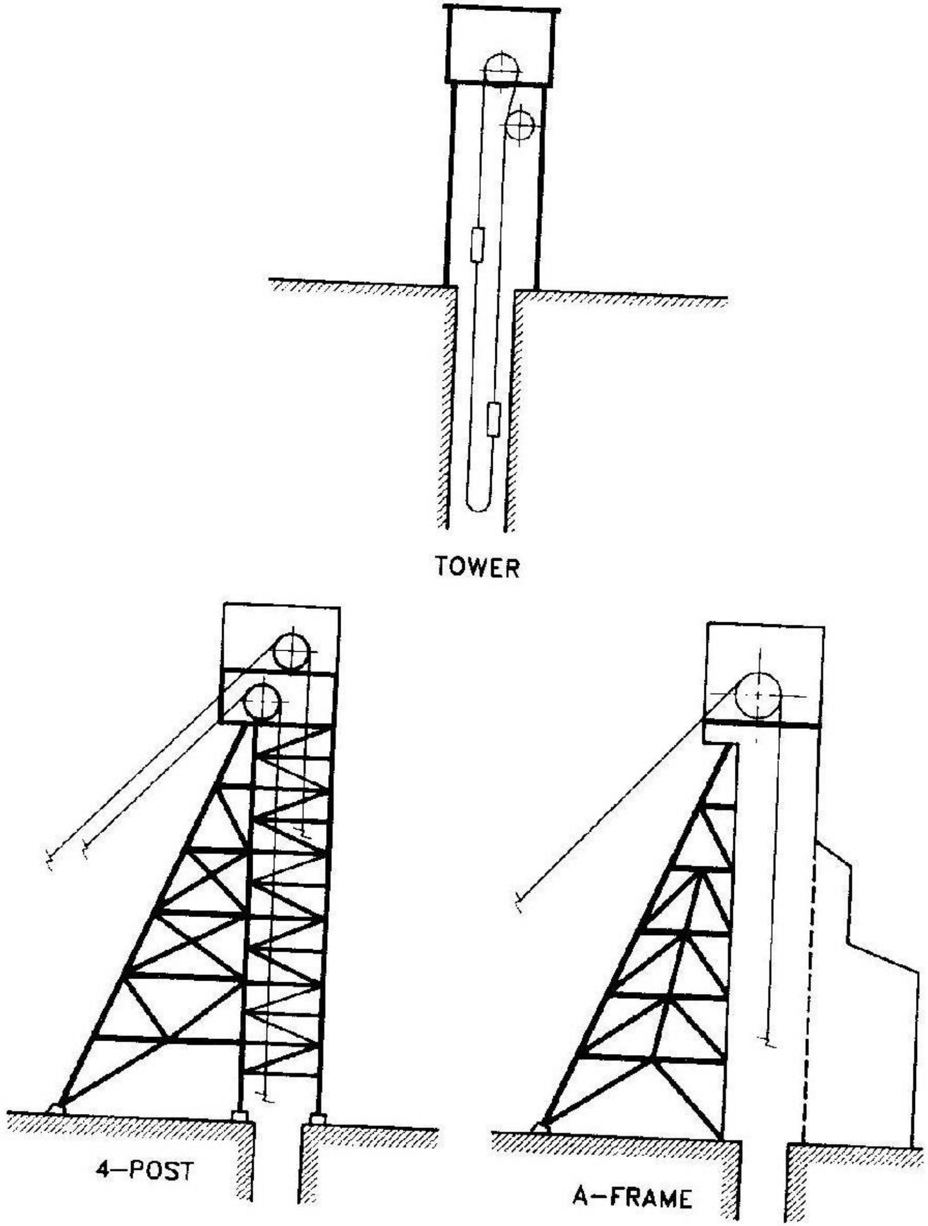

Fig. 17.5.15. Types of headframe structures.

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Head Frame II

*Head frame is installed over shaft to support sheave wheel over which hoisting rope passes for raising & lowering conveyance.

*Some time entire winder is mounted over head frame.

*Construction of head frame is also necessary to allow dumping of hoisted material above ground.

*Structurally there are two type of head frame.

*Head frame with back leg (A type head frame)

*Four or six post tower type head frame.

Construction Material Head Frame I

*Head frame can be constructed by timber, steel, concrete.

*Present mines are designed with high capacity winder from deep shaft, hence use of timber for head frame construction has been discontinued.

*All modern head frame are made of steel or concrete.

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Concrete Head Frame I

Advantages

*Effective use can be made of space within structure for tower mounted friction winder.

*Maintenance cost is low.

*High dampening effect of massive concrete structure on shock & vibration.

*Simplicity of shaft collar layout & interconnection with service building.

*Durability & resistance to corrosion.

Disadvantages

* Lengthy construction time.

Concrete Head Frame II

*Higher capital cost in comparison of steel head frame.

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Steel Head Frame I

Advantages

*Lighter foundation load.

*Adaptability to mine changes.

*Salvage cost.

*Capital cost saving at remote area.

*Can be fabricated & transported to site for assembly.

*Speedy construction.

Disadvantages

*Prone to corrosion & thus not suitable for production head frame for corrosive ore (salt, sulphates) unless head frame steel is treated.

Head Frame Design I

*Head frame is to be designed in relation with shaft & winding system, room etc.

*Design of head frame should include following consideration.

*Load.

*Foundation.

*Mine service.

*Sinking provision.

*Equipment monitoring facilities.

*Conveyance handling & rope handling.

*Heating & ventilation.

Head Frame Design II

*Local mining regulation.

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Head Frame Loading I

*All head frame should be designed to with stand a combination of load.

*Dead load consisting of weight of head frame , sheave wheel, ore bins & contents.

*Live load from hoisting at maximum capacity.

*Breaking load of ropes when conveyance is stopped by crash beam or if it is jammed in shaft.

*Breaking stress introduced by an over wind are usually transferred to structured system, often through an arrester gear.

*Supporting structure must be critically analyze with a through understanding of arrester mechanism.

Head Frame Loading II

*Wind load, intensity of which depends on location, height & shape of structure.

*Snow load.

*Temperature & seismic stress taken from published data relating to geographic area of shaft location.

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Head Frame Foundation I

*Foundation should be capable of with standing & absorbing stress imposed by structure & associated load.

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Head Frame Mine Service I

*Design of head frame must suit surface layout for providing essential mine services.

*Head frame should be located close to shaft coller house, waiting areas, lamp room, first aid room, winder house, maintenance shop, changing house, administrative office.

*Design must take in respect of shaft, position of sheave wheel.

*Arrangement of skip dumping are in relation to ore & waste & handling facilities.

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Head Frame Mine Service II-A

*Clearance required for installation, maintenance & removal of conveyance

*Lifting equipment required for maintenance of conveyance.

*Access required for maintenance of conveyance, dump ore etc.

*Lifting equipment required for maintenance of conveyance

*Access & anchorage facility to support end hoist rope.

*Sub- collar loading access double deck cage,

Head Frame Mine Service II-B

*Provision of hoisting underground machines for servicing / repairing etc.

Head Frame Mine Service III

*Supporting facility at predetermined elevation for hoist or balance ropes, conveyance & counter weight during installation & changes.

*Clearance for installation of conveyance arrester gear.

*Provision for handling shaft ventilation air & for sealing machining room.

*Arrangement for MG or power convertor close to winding motor.

*Lifting equipment access during main & tail rope changing.

Head Frame Mine Service IV

*Support, accommodation, inspection, removal & replacement of facilities for rope guide.

*Ventilation to specific machines & area, controlling moisture, dust & temperature.

*Accommodation of sinking arrangement.

*Accommodation for all services air, water, communication, power line to shaft & head frame equipment.

*Accommodation of stair cases & elevator.

*Lighting especially at unloading point, ore bin discharge, head sheave pulley, stairs, lift, cage landing area & general lighting at other area.

Sinking Head Frame I

*Shaft sinking can be done by using temporary sinking head frame & sinking winder.

*Normally contractor brings sinking head frame & sinking winder.

*This head frame may be made out of wood or steel structure in modular form.

*Temporary steel head frame is fabricated from hollow structure steel, which combine strength with light weight.

*Light weight head frame is easy to erect & dismantle.

*Head frame of this type are portable & has high resale value.

Sinking Head Frame II

*Height of temporary head frame depends up on method of rock disposal & size of bucket.

*Sinking head frame embody for dumping bucket or skip.

*System should protect personal in surface & inside shaft from falling piece of rock while dumping.

*Head frame design should allow minimum man power to dump bucket & removal of broken rock.

*Temporary head frame should accommodate an adequate size strong bin & sufficient clearance for rock disposal vehicle

Sinking Head Frame III

*Sinking head frame should have provision of fixing removable trap door to keep shaft mouth allays closed, except for movement of conveyance,.

*Sinking head frame is fitted with bell hammer to be used as signal system with shaft face.

*Head frame platform should have provision for ventilation fan for ventilating underground.

*Head frame platform should have shaft survey facility.

*Head frame platform should have provision of concrete pouring for lining & other purposes,

Sinking Head Frame IV

*Current trend favour installation of a permanent head frame for sinking operation.

*It is more economical & time saving.

*Permanent main head frame skeleton can be designed to facilitate erection & installation of sinking sheave at a suitable elevation.

*Use of permanent structure as a sinking head frame can make permanent hoisting facilities available at earlier date following completion of shaft sinking.

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Head Frame Equipment I

*Over travel at shaft extremities (sump, head frame) has be monitored & prevented.

*Safety device be installed in head frame.

*Track limit switches are installed, above limit over which conveyance is not supposed to move.

*Once, conveyance cross this limit, power to winder is cut off & emergency brake to be applied.

*Head frame arrester are designed to retard ascending conveyance from full speed to halt.

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Equipment Handling Head Frame I

*In design & layout of a head frame, provision must be made for handling & designing of conveyance, hoisting rope & rope guide.

*Tower mounted winder head frame should have special rope handling & conveyance arresting mechanism to facilitate rope changing.

*Tower mounted friction winder should have over head travelling crane for lowering & hoisting winding room equipment like winder, gear box, motor etc.

Equipment Handling Head Frame II

*In case of rope guide, a lubricant discharge arrangement is placed on rope guide to lubricate rope on regular basis.

*Head frame should have lighting arrester & arrangement should be available to check ground resistance of lighting arrester.

*Toll head frame must be fitted with aviation light.

*In case of friction winder, automatic conveyance synchronizer is located at head frame.

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Head Frame Ventilation I

*In warm climate, steel head frame leaves open lot of space & there is no hindrance for ventilation air to get into mine.

*In case of RCC tower mounted head frame sufficient opening has to be kept for ventilation air to get in mines.

*In cold climate, head frame is an enclosed structure with adequate provision of heating system.

*In tower mounted friction winder, winding room is located at top of tower.

Head Frame Ventilation II

*It is necessary to provide enclosure to accommodate winder, motor, controls etc in head feame.

*Head frame should have sufficient space, ventilation system, cooling arrangement for hot countries by supplying filtered air for cooling.

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Head Frame Height I

*Height of head frame depends on following.

*Clearance for loading facility under storage bin.

*Height of head frame should ensure that resultant force from head frame falls with in back leg post.

*Providing minimum clearance for conveyance over travel beyond its normal travel to head frame sheave.

*Minimum clearance for over run distance is principle design consideration.

*Free run distance in head frame depends on legal requirement.s

*Operating allowance.

Head Frame Height II

*Rope stretch

*Stopping allowance,

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Rope Installation I

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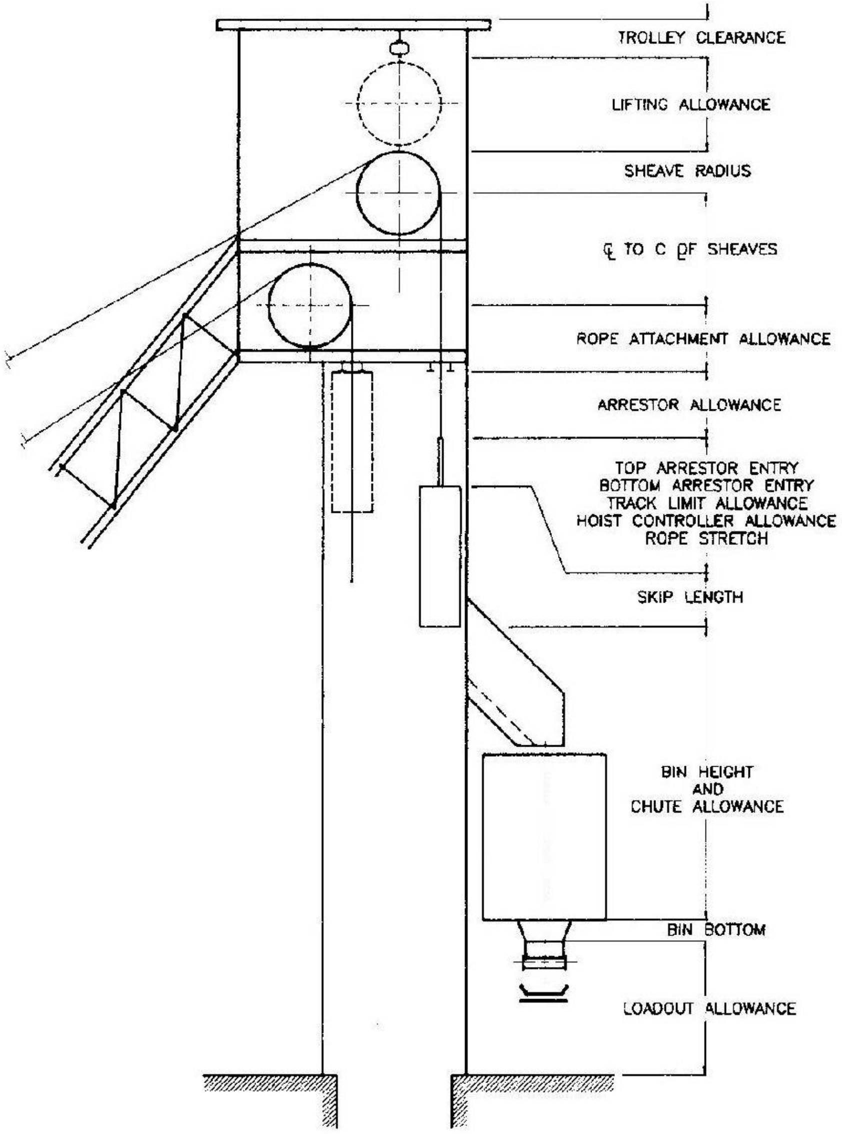

Fig. 17.5.16. Vertical general arrangement of headframe.

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Head Frame Operating Allowance I

*An operating allowance is required to allow for variation in repeatability of final stop position of conveyance in head frame.

*In malty level hoisting with drum hoist, this tolerance compensates for difference in coiling & rope stretch.

*In case of friction winder, allowance compensates for creep during re- synchronization & also allows a reasonable gap for rope adjustment due to permanent rope stretch.

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Head Frame Rope Stretch Allowance I

*This is included to avoid contact with switch when empty conveyance are hoisted.

*Rope stretch allowance generally includes additional 0.3m margin as well as extra travel length during brake time & because effect of de acceleration.

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Head Frame Stoppage allowance II

* A stopping allowance is to ensure that clearance between top of conveyance & first obstruction in head frame.

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Reference I

*SME Mining Engineering Hand Book, Volume 2, 2nd Edition. Page 1674-1677

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